Decimal, Binary and Hexadecimal Number Systems

EET 100D: Introduction to Computers and Networking
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Positional Number System

- Assume there is a three digit number that includes two fives and a three. What are the possible numbers?
  - 553, 535, 355
- Can you know the value without the order of the numbers?
  - No!
- What does this mean? It means that any value expressed in numbers (digits) not only depends on the digits but the order or position thereof;
- Number system using 10/2/16 distinct symbols and position (column weights) is called the decimal/binary/hexadecimal system:
  - Decimal: human-related math (humans have 10 fingers);
  - Binary: machine/digital representations (machines have two distinct states);
  - Hexadecimal: human understanding of machines

Decimal System

How do we humans count:

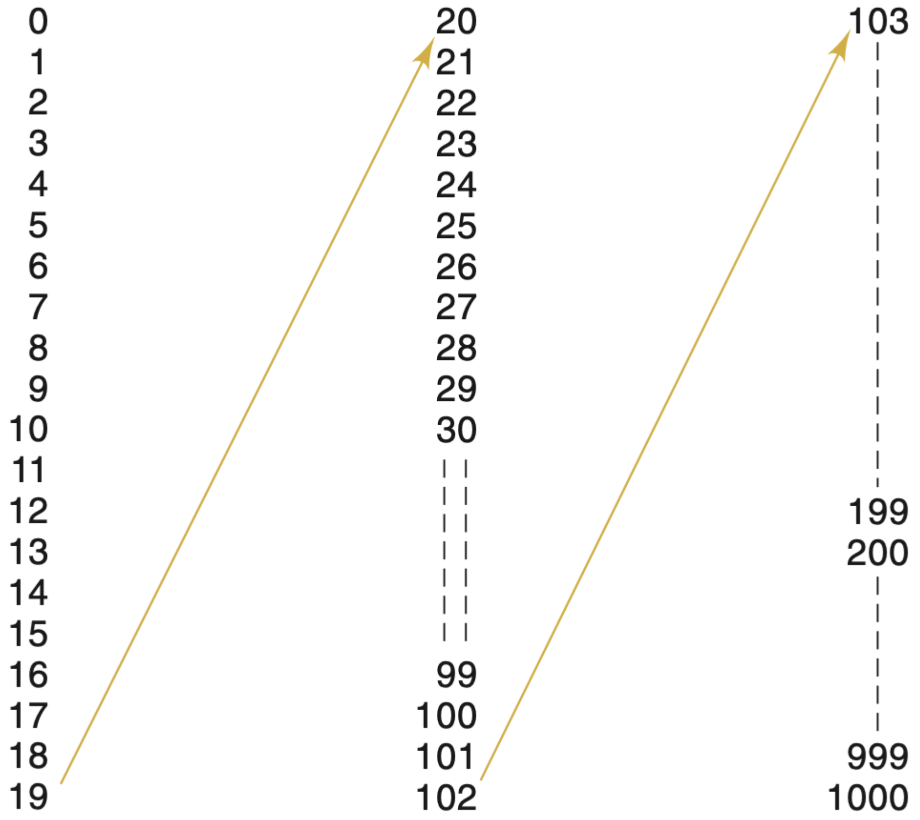
- 10 distinct numerals, symbols, digits or human fingers: 0, 1, …, 9;
- Cout from 0 to 9 with the 10 distinct numerals;
- Counting 1 more from 9, when the 10 distinct numerals are exhausted, we use 10, where the *position* (the "tens" position) of the numeral 1 means that it represents 10;
- Then we increment the "unit" position and count 10, 11, …, 19;
- When the unit position exhausts the 10 distinct numerals, we increment 1 in the tens position and count 20, 21, …, 29.

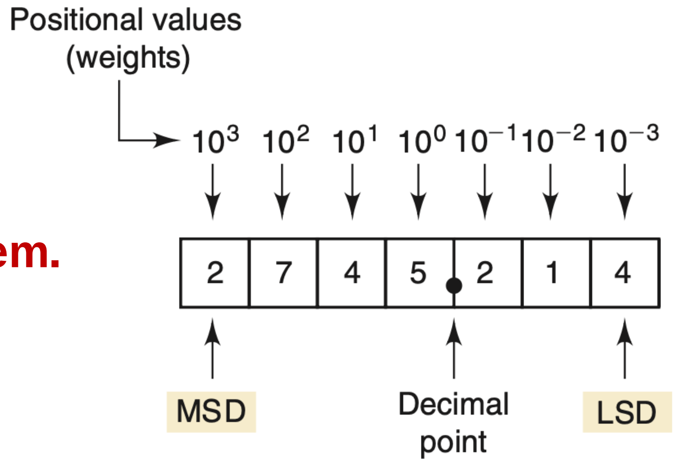
The value of each numeral in a number is determined by the numeral itself and its position:

$$2745.214$$
$$= 2 \times 10^3 + 7 \times 10^2 + 4 \times 10^1 + 5 \times 10^0 + 2 \times 10^{-1} + 1 \times 10^{-2} + 4 \times 10^{-3}$$

With 2 digits, can count 100 (= 10^2) numbers from 0 to 99 (= $10^2 - 1$):

- With N digits, can count 10^N numbers from 0 to $10^N - 1$.





This 10-based counting, quantity-representing system is called the decimal system.

Binary System

How do computing machines count:

- 2 distinct numerals, symbols, digits or machine states: 0, 1;
- Cout from 0 to 1 with the 2 distinct numerals;
- Counting 1 more from 1, when the 2 distinct numerals are exha we use 10, where the *position* of the numeral 1 means that it re 2;
- Then we increment the “unit” position and count 10, 11;
- When the unit position exhausts the 2 distinct numerals, we inc in the 2nd position and find that the 2nd position has exhausted numerals, so we use the 3rd position and count 100.

The value of each numeral in a number is determined by the num and its position:

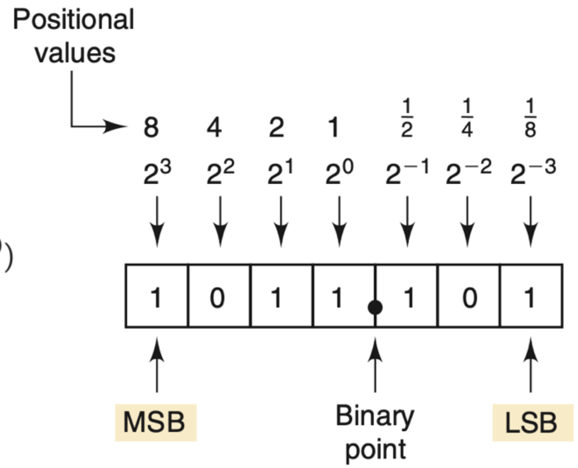
With 2 digits, can count 4 (= 2^2) numbers from 0 to 11 (= $2^2 - 1 = 3$):

- With N digits, can count 2^N numbers from 0 to $2^N - 1$.

Weights →	$2^3=8$	$2^2=4$	$2^1=2$	$2^0=1$	Decimal equivalent
	0	0	0	0	0
	0	0	0	1	1
	0	0	1	0	2
	0	0	1	1	3
	0	1	0	0	4
	0	1	0	1	5
	0	1	1	0	6
	0	1	1	1	7
	1	0	0	0	8
	1	0	0	1	9
	1	0	1	0	10
	1	0	1	1	11
	1	1	0	0	12
	1	1	0	1	13
	1	1	1	0	14
	1	1	1	1	15
				LSB	

This 2-based counting, quantity-representing system is called the binary system.

$$1011.101_2 = (1 \times 2^3) + (0 \times 2^2) + (1 \times 2^1) + (1 \times 2^0) + (1 \times 2^{-1}) + (0 \times 2^{-2}) + (1 \times 2^{-3})$$
$$= 8 + 0 + 2 + 1 + 0.5 + 0 + 0.125$$
$$= 11.625_{10}$$



Hexadecimal System

16 distinct numerals:

- The numbers 0 through 9 and the alphabetic characters A through F;
- Same counting sequence: 1…9, A, B, C, D, E, F, 10, … 1F, … FF, 100, …;
- Same position-numeral combination to represent quantities;

With 2 digits, can count 256 (= 16^2) numbers from 0 to 255 (= $16^2 - 1 = 255$):

- With N digits, can count 2^N numbers from 0 to $2^N - 1$.

16 = 2^4 makes it easy to represent binary numbers with hexadecimal numbers:

- Group 4 bits at a time starting from LSD: $1001\ 0110\ 0000\ 1110_2$
- Use the table to find: $1001\ 0110\ 0000\ 1110_2 = (960E)_{16}$ = 960Eh ("h" means hex or hexadecimal).

16^4	16^3	16^2	16^1	16^0	16^{-1}	16^{-2}	16^{-3}	16^{-4}

Hexadecimal point

0	0	0
1	1	1
2	2	10
3	3	11
4	4	100
5	5	101
6	6	110
7	7	111
8	8	1000
9	9	1001
A	10	1010
B	11	1011
C	12	1100
D	13	1101
E	14	1110
F	15	1111
10	16	10000
11	17	10001
12	18	10010
13	19	10011
14	20	10100
15	21	10101
16	22	10110
17	23	10111
18	24	11000
19	25	11001
1A	26	11010
1B	27	11011
1C	28	11100
1D	29	11101
1E	30	11110
1F	31	11111
20	32	100000